

Weak Form of Market Efficiency: Evidence from Nepalese Stock Market

Surya Bahadur G.C.*

Abstract

This paper examines the weak form market efficiency of Nepalese stock market by using 1970 daily observations from 2003 to 2009 of the general NEPSE index and seven different sector-wise indices. The study employs five different tests of random walk: autocorrelation test, runs test, unit root tests (ADF, PP, and KPSS), variance ratio test and autoregressive conditional heteroskedasticity test. The results of the tests are in broad agreement, conclusively rejecting the presence of random walk in daily returns of the eight stock market indices. The serial correlation tests and the runs tests both revealed that the successive price changes are not random and are serially dependent. Similarly, the unit root tests conclude that unit roots, as necessary conditions for a random walk, are absent from all of the return series. Finally, the results of variance ratio and GARCH (1,1) procedures convincingly reject the presence of random walks in the index return series implying that the Nepalese stock market is not efficient in the weak form.

Keywords: Random walk, Weak form market efficiency, Nepalese stock market, Unit root test, Variance ratio test, Autocorrelation test, Runs test, GARCH

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1. Introduction

The capital market plays various roles in an economy. It acts as an intermediary between surplus units and deficit units of the economy and facilitates savings into investments. By also providing liquidity of these investments, the capital market ensures optimum allocation of resources. A fundamental question concerning capital markets is their efficiency. Testing of the financial markets efficiency is an important issue for investors and policymakers dealing with emerging equity markets. Efficiency has deep implications

for asset pricing, risk and returns on the exchange. According to Samuelson (1965) and Fama (1970), under the efficient market hypothesis (EMH), stock market prices must always show a full reflection of all available and relevant information and should follow a random walk process. Successive stock price changes (i.e. the stock returns) are therefore independently and identically distributed. Fama (1970) categorizes the three types of efficient markets as weak-form, semi-strong-form, and strong-form efficient if the set of information includes past prices

* Mr. G.C. is Lecturer, School of Business, Pokhara University

Surya Bahadur G.C.

and returns only, all public information, and any information public as well as private, respectively. The implication here is that all markets can be weak-form but the reverse cannot be the case.

The weak form of EMH asserts that studying the behavior of historical prices cannot earn abnormal returns. The implication of weak-form efficiency is the random walk hypothesis (RWH), which indicates that successive price changes are random and serially independent. Although there are an abundance of empirical studies concerning testing the RWH (Liu and He (1991), Huang (1995), Poshakwale (1996), Islam and Khaled (2005), etc.), the interest in the market efficiency still remains in academicians and practitioners. It is usually believed that the markets in developing and less developed countries are not efficient in semi-strong form or strong form.

The increasing prominence of the stock market in Nepal is one of the most striking features of financial development over the last decade. Securities Exchange Center Limited was established in 1976 to facilitate and promote the growth of capital markets. As part of the economic liberalization program in Nepal, it was converted into Nepal Stock Exchange (NEPSE), under a program to reform capital markets in 1993. In January 1994 NEPSE opened public share trading based on the open outcry system with twenty five brokers and

three market-makers. Since then, NEPSE had brought about a number of changes in order to upgrade itself and provide efficient and reliable services.

The recent reforms have focused on enhancing institutional development. To ensure transparency and dynamism in stock transactions, the open voice bidding system has been replaced by computerized transactions. The NEPSE adopted the Automated Trading System (ATS) instead of open outcry trading system from August 24, 2007, permitting a high degree of price transparency and real-time price quotations. The ATS has started on-line trading through Wide Area Network (WAN). Now, the brokers have been able to trade via computers either on the floor of the exchange or from their offices. As NEPSE became more systematic, it has implemented index-based circuit breakers with effect from 21 September 2007. In addition to the circuit breakers, price range is also applicable on individual securities to stabilize the volatility of the stock market. The evolving capital market is starting to attract small investors, which is a good sign for its further consolidation.

The academic literature on market efficiency in Nepal is scanty. Baral and Shrestha (2006) and Pradhan and Upadhyay (2006) conducted tests on weak form of efficiency at NEPSE using serial correlation and run test and found evidence of market inefficiency in weak form.

Using regression model with dummies, K.C. and Joshi (2005) find persistent evidence of day-of-the-week anomaly but disappearing holiday effect, turn-of-the-month effect and time of-the-month effect. Their results indicate that the Nepalese stock market is not efficient in weak form with regard to the day-of-the-week anomaly but weakly efficient with respect to the other anomalies. Bhattarai and Joshi (2007) investigated whether or not the Nepalese stock market is efficient in weak form with respect to economically neutral behavioural variables using simple OLS technique with White's heteroskedasticity-corrected standard errors to test the relationship between stock returns and economically neutral behavioural variables represented by weather (cloud cover and temperature) and biorhythms (seasonal affective disorder). The findings indicate the existence of weak-form efficiency in the market for "temperature" and "seasonal affective disorder" but not for the "cloud cover".

However, the Nepalese stock market has not been deeply researched in terms of its degree of efficiency using more robust approaches or methods. The study of market efficiency including the data after the reforms and introduction of computerized trading system has not been undertaken. Theoretically, we expect a young emerging capital market such as the NEPSE to be weakly efficient at best. The objective of this paper is to determine

the degree of efficiency of the NEPSE in the face of on going developments. The study findings are expected to serve the investors for gaining profit, providing an evidence to be added for international evidences, and will enable the policymakers to improve the level of efficiency. The presence or absence of random walk in the price generation process in a stock market is evaluated using stock market indices. The study is made comprehensive by including daily return series of eight stock market indices for a longer time framework.

2. Brief Literature Survey

Although, the first tests of market efficiency were reported as early as 1900, the classification of efficiency into three levels did not emerge until 1959 since it was not until the fifties that research in the area developed systematically. The concept of market efficiency had been anticipated at the beginning of the century in the dissertation submitted by Bachelier (1900). He stated that past, present and even discounted future events are reflected in market price, but often show no apparent relation to price changes. The problem of the optimal search procedure for finding a drunk left in a middle of a field, expected to stagger randomly was discussed early in the century by Pearson (1905). Kendall (1953) examined 22 UK stock and commodity price series and concluded that the data behaved like wandering series. In the 1960s and early 1970s, the

controversy focused on the extent to which successive changes in prices of the stocks were independent of each other or whether stock prices followed a random walk. The near zero serial correlation of price changes came to be labeled as the random walk model. In their papers, Samuelson (1965) and Fama (1970) provided comprehensive studies on the issue and claimed to have found strong evidence in support for the Random Walk Hypothesis.

Other studies in the US with similar findings included those of Sharpe (1966), Friend et al. (1970), and Williamson (1972). Fisher (1996) also suggested autocorrelations of monthly returns. The studies of Granger and Morgenstern (1963), and Solnik (1973) supported the random walk model. Cooper (1982) studied 36 world stock markets to examine the validity of the random walk hypothesis. With respect to US and UK, the random walk hypothesis was accepted while it was rejected in all other markets. Harvey (1995) and Khababa (1998) found that the market of developing and less developed countries are not efficient in weak form.

The findings of market efficiency tests (mostly weak-form efficiency tests) on emerging markets are rather varied. Some researchers find evidence in favour of weak-form efficiency (Cheung and Coutts, 2001; and Abrosimova and Linowski, 2002), but some other researchers find evidence of

predictability of stock prices (e.g. Lee et al., 2001 and Smith et al., 2002). Alam, Hasan, and Kadapakkam (1999) examine five Asian markets (Bangladesh, Hong Kong, Malaysia, Sri Lanka, and Taiwan) and conclude that all the index returns follow a random walk with the exception of Sri Lanka. Darrat and Zhong (2000) and Poshakwale (2002) reject the RWH for the Chinese and Indian stock markets, respectively. Hoque, Kim, and Pyun (2007) test the RWH for eight emerging markets in Asia using Wright's (2000) rank and sign VR tests and find that stock prices of most Asian developing countries do not follow a random walk with the possible exceptions of Taiwan and Korea. Worthington and Higgs (2006) examine the weak-form market efficiency of twenty-seven emerging markets using daily market returns and tested for random walks using serial correlation coefficient and runs tests, Augmented Dickey-Fuller (ADF), Phillips-Perron (PP) and Kwiatkowski, Phillips, Schmidt and Shin (KPSS) unit root tests and conclude that most emerging markets are weak-form inefficient.

3. Empirical Methodology

3.1 Data

The sample includes total 1970 daily observations of different NEPSE daily general price index for the total sample period from 2003 to 2009 A.D. NEPSE Index was a natural choice for including in this study, as it is the most popular market index and widely

used by market players for benchmarking. The index is composed of all the listed equities on the market. In addition to the NEPSE Index, seven other industry-wise indices available, which are widely used as the indicator of the performance of the different sectors of the stock market in Nepal are used in the study. The daily closing prices of respective indices are used as the source data. The span of time period in sampling of data was restricted due to the availability of data. The required data were obtained from the official database of the NEPSE.

The return is calculated as the logarithmic difference between two consecutive prices in a series, yielding continuously compounded returns. The reasons to take logarithm returns are justified by both theoretically and empirically. Theoretically, logarithmic returns are analytically more tractable when linking returns over longer intervals. Empirically, logarithmic returns are more likely to be normally distributed which is a prior condition of standard statistical techniques. Daily market returns (R_t) are calculated from the daily price indices such as follows:

$$R_t = \ln (PI_t / PI_{t-1}) \quad (1)$$

Where,

R_t = market return at period t ;

PI_t = price index at period t ;

PI_{t-1} = the price index at period $t-1$ and

$\ln$ = natural log.

3.2 Research Methods

Many statistical tests for random walks (or EMH) have been used in the literature. The different methods for testing random walk used in the study have been described briefly below:

3.2.1 The Q-Statistic

Box & Pierce (1970) developed Q-statistic as an alternative to various hypotheses of autocorrelation with different time lags. Box-Pierce Q-statistic is a linear combination of squared autocorrelations with all the weights set identically equal to unity. It is defined as:

$$Q = T \sum_{k=1}^m \hat{t}_k^2 \quad (2)$$

Where, $\hat{t}_k$ = autocorrelation with k lags, T = sample size and m = maximum lag length. The Q -statistic is asymptotically distributed as a χ^2_m under the null hypothesis that all m autocorrelation coefficients are zero. As for any joint hypothesis test, only one autocorrelation coefficient needs to be statistically significant for the test to result in a rejection. A variant of the Box-Pierce test, having better small sample properties known as Ljung-Box (1978) statistics has been developed. The Q-statistic at lag k is a test statistic for the null hypothesis that there is no autocorrelation up to order k and is computed as:

$$Q^* = T(T+2) \sum_{k=1}^m \frac{\hat{t}_k^2}{T-k} \sim c_m^2 \quad (3)$$

This statistic is very useful as a portmanteau test of linear dependence in time series.

3.2.2 The Runs Test

The runs test, is a non-parametric test in which the number of sequences of consecutive positive and negative returns is tabulated and compared against its sampling distribution under the random walk hypothesis. A run is defined as the repeated occurrence of the same value or category of a variable. It is indexed by two parameters, which are the type of the run and the length. Stock price runs can be positive, negative, or have no change. The length is how often a run type occurs in succession. Under the null hypothesis that successive outcomes are independent, the total expected number of runs is distributed as normal with the following mean:

$$m = \frac{N(N+1) - \sum_{i=1}^3 n_i^2}{N} \quad (4)$$

and the following standard deviation:

$$a_m = \left[\frac{\sum_{i=1}^3 [S_{i=1}^3 [S_{i=1}^3 n_i^2 N(N+1)] - 2N(S_{i=1}^3 n_i - N_i^3)]}{N^2(N-1)} \right]^{\frac{1}{2}} \quad (5)$$

where n_i is the number of runs of type i . The

test for serial dependence is carried out by comparing the actual number of runs, a_r in the price series, to the expected number μ . The null proposition is:

$$H_0: E(runs) = \mu \quad (6)$$

3.2.3 Unit Root Tests

Usually time series analysis considers stationary time series in empirical studies. A series is said to be stationary if the mean and auto-covariance of the series does not depends on time. In order to examine whether each variable's time series is integrated and has a unit root, the study has considered three widely used popular unit root tests, namely, the Augmented Dickey-Fuller (ADF) test (1979), the Phillips-Peron (PP) test (1988), and the Kwiatkowski, Phillips, Schmidt and Shin (KPSS) test (1992). The presence of unit root implies non-stationary time series exhibiting random walk. Absence of unit root indicates that the time series is stationary and doesn't follow a random walk.

The equation of unit root test is,

$$R_t = \alpha + \rho R_{t-1} + e_t \quad (7)$$

Where e_t is the error term with zero mean, constant variance and α is the intercept. If $\rho = 1$, means unit root is present and the series are random walk. The popular ADF unit root test of the null hypothesis of non-stationary (exhibiting a random walk) expressed as:

$$\Delta R_{kt} = \alpha_0 + \alpha_1 t + r_0 R_{kt-1} + \sum_{k=1}^q \rho_k \Delta R_{kt-k} + e_{kt} \quad (8)$$

Where, R_{kt} denotes the return for the k -th market at time t and $\Delta R_{kt} = R_{kt} - R_{kt-1}$, r are coefficients to be estimated, q is the number of lagged terms, t is the trend term, α_1 is the estimated coefficient for the trend, α_0 is the constant, and ε is white noise. MacKinnon's critical values are used in order to determine the significance of the test statistic associated with ρ_0 . The unit root tests the null hypothesis $H_0: r = 1$ against the one-sided alternative $H_1: r < 1$. The null hypothesis of a unit root is rejected in favour of the stationary alternative in each case if the test statistic is more negative than the critical value.

The PP incorporates an alternative (nonparametric) method of controlling for serial correlation when testing for a unit root by estimating the non-augmented Dickey-Fuller test equation and modifying the test statistic so that its asymptotic distribution is unaffected by serial correlation. Finally, the KPSS uses a similar (though parametric) autocorrelation correction to the PP but assumes that the observed time series can be decomposed into the sum of a deterministic trend, a random walk with zero variance and a stationary error term. It thus tests the null hypothesis of trend stationarity corresponding to the hypothesis that the variance of the

random walk equals zero.

It is well known that ADF unit root tests fail to reject the null hypothesis of a unit root for many time series, and that allowing for error autocorrelation using the PP test does not necessarily improve these results. However, the KPSS test complements the standard unit root tests since it can distinguish between the logarithm of the prices that appear to be stationary, those that appear to have a unit root, and those that are not sufficiently informative to be sure whether they are either.

3.2.4 The GARCH (1, 1) Model

Engle (1982) and Bollerslev (1986) independently introduced the autoregressive conditional heteroscedasticity (ARCH) and the generalized ARCH (GARCH) models, which specifically allows for a time variant conditional variance and nonlinearities in the generating mechanism. The basic GARCH (1, 1) given below is estimated in this study as the lag order (1, 1) model is sufficient to capture all of the volatility clustering that is present from the data.

The GARCH(1, 1) model by Bollerslev (1986) is based on the assumption that forecasts of time varying variance depend on the lagged variance of the asset. An unexpected increase or decrease in returns at time t will generate an increase in the expected variability in the next period. The basic model GARCH (1, 1) can be expressed as:

$$h_t = w + a_1 e_{t-1}^3 + b_1 h_{t-1} \quad (9)$$

The variance (h_t) is a function of an intercept (w), a shock from the prior period (a) and the variance from last period (b). The mean equation for the model is:

$$R_t = m + qR_{t-1} + e_t \quad (10)$$

Where; $w > 0$, $a \geq 0$, $b \geq 0$. The GARCH (1, 1) is weakly stationary if $(a+b) < 1$. If $(a+b)$ is very close to one, it shows high persistence in volatility clustering and implies inefficiency on the market and if $\theta = 0$ and the error term is *i.i.d* in the mean equation we accept weak form EMH otherwise we reject the hypothesis.

3.2.5 Variance Ratio Test

The robust single variance ratio test developed by Lo & MacKinlay (1988) is used on the eight market indices selected for the study. As shown by Lo and MacKinlay, the variance ratio statistic is derived from the assumption of linear relations in observation interval regarding the variance of increments. If a series follows a random walk process, the variance of a q th differenced variable is q times as large as the first-differenced variable. For a series partitioned into equally spaced intervals and characterized by random walks, one q th of the variance of $(R_t - R_{t-q})$ is expected to be the same as the variance of $(R_t - R_{t-1})$:

$$\text{Var}(R_t - R_{t-q}) = q \text{Var}(R_t - R_{t-1}) \quad (11)$$

where q is any positive integer. The variance

ratio is then denoted by:

$$VR(q) = \frac{1/q \text{Var}(R_t - R_{t-q}) - \text{Var}[R_t - (q)]}{\text{Var}(R_t - R_{t-1}) - q \cdot \text{Var}[R_t]} = 1 + 2 \sum_{k=1}^{q-1} \left(1 - \frac{k}{q}\right) r(k) \quad (12)$$

such that under the null hypothesis, $VR(q) = 1$ for all q . The $r(k)$ is the k th order autocorrelation coefficient of $\{R_t\}$. The null hypothesis is stated as H_0 : The variance ratio at lag q is defined as the ratio of the variance of the q -period return to the variance of the one-period return divided by q , which is unity under the random walk hypothesis. The random walk null hypothesis is rejected if any one of the estimated variance ratio is significantly different from one.

4. Empirical Results

4.1 Autocorrelation Test

The autocorrelation function test using Ljung-Box methodology is examined to identify the degree of autocorrelation in the stock indices return series. It measures the correlation between the current and lagged observations of the time series of stock returns. Table-1 presents the tests of independence using different lags. All of the null hypotheses of no serial correlation for the eight different market indices are rejected at high significance level. The significance of the autocorrelation coefficient indicates that the null hypothesis of weak-form market efficiency may be rejected and we may infer all industry stocks to be weak-form inefficient.

Table 1: Output of Autocorrelation Tests

CBI				DBI			FINI			HOTI		
Lag	AC	Q-Stat	Prob	AC	Q-Stat	Prob	AC	Q-Stat	Prob	AC	Q-Stat	Prob
1	0.215	90.836	0.000	-0.247	120.410	0.000	-0.384	291.040	0.000	0.071	9.962	0.002
2	-0.018	91.455	0.000	0.015	120.840	0.000	0.013	291.350	0.000	0.054	15.805	0.000
3	-0.014	91.867	0.000	0.044	124.740	0.000	0.020	292.160	0.000	0.058	22.406	0.000
4	-0.003	91.881	0.000	0.019	125.430	0.000	0.005	292.210	0.000	0.070	32.182	0.000
5	-0.002	91.891	0.000	0.015	125.900	0.000	0.021	293.080	0.000	0.040	35.379	0.000
10	-0.019	92.822	0.000	0.016	131.670	0.000	0.008	294.290	0.000	0.013	36.564	0.000
15	0.028	100.710	0.000	-0.008	132.910	0.000	0.015	295.950	0.000	0.000	37.868	0.001

INSI				MFGI			TRDI			NEPSEI		
Lag	AC	Q-Stat	Prob	AC	Q-Stat	Prob	AC	Q-Stat	Prob	AC	Q-Stat	Prob
1	-0.482	458.240	0.000	-0.250	123.030	0.000	-0.477	448.270	0.000	0.270	143.820	0.000
2	0.003	458.260	0.000	0.000	123.030	0.000	0.001	448.280	0.000	0.004	143.850	0.000
3	-0.004	458.290	0.000	0.003	123.050	0.000	0.000	448.280	0.000	0.025	145.100	0.000
4	0.006	458.360	0.000	-0.158	172.690	0.000	0.000	448.280	0.000	0.011	145.320	0.000
5	-0.003	458.380	0.000	-0.001	172.690	0.000	0.000	448.280	0.000	0.022	146.270	0.000
10	0.002	458.610	0.000	-0.002	174.800	0.000	0.000	448.280	0.000	-0.006	150.100	0.000
15	-0.003	458.770	0.000	-0.003	174.970	0.000	0.000	448.280	0.000	0.036	169.900	0.000

Notes: CBI – commercial bank index, DBI – development bank index, FINI – Finance company index, HOTI – hotel sector index, INSI – insurance sector index, MFGI – Manufacturing sector index, TRDI – trading company index, and NEPSEI – NEPSE index.

The relevant critical values are from a χ^2 distribution. Clearly, in all cases the joint null hypothesis that all of the first k lag autocorrelation coefficients are zero are rejected. It is clearly evident from the result that the series are very persistent, with autocorrelation functions that die away very slowly. The penultimate column of the table gives the statistic resulting from a Ljung-Box test which rejects the null hypothesis of no autocorrelation. The non-zero autocorrelation of the series associated with the Ljung-Box Q-statistics, which are jointly significant at 1% level suggest that the return series doesn't follow a random walk. The autocorrelations

seem to persist. The significance of autocorrelation coefficient reveals that the null hypothesis of weak-form market efficiency is rejected. Hence, the NEPSE seems to be inefficient in the weak form.

4.2 The Runs Test

The Runs Test has been used to see if a index series is random. The runs test is another approach to test and detect statistical dependencies or randomness which may not be detected by the auto-correlation tests. The null hypothesis of the test is that the observed series is a random series. Runs Test is a nonparametric test because no assumption

is made about population distribution parameters. This test is used for determining if the order of responses above or below a specified value is random. A run is a set of consecutive observations that are all either less than or greater than a specified value.

The number of runs is computed as a sequence of price changes of the same sign. When the expected number of run is significantly different from the observed number of runs, the test rejects the null hypothesis that the daily index return series are random.

Table 2: Output of the Runs Test

	Cases < Median	Cases >= Test Median	Total Cases	Number of Runs	Z-Value	p-value
NEPSEI	527	1443	1970	581	-11.045	0.000
CBI	555	1415	1970	599	-11.098	0.000
MFGI	93	1877	1970	163	-3.822	0.000
HOTI	170	1800	1970	293	-2.67	0.008
TRDI	52	1918	1970	89	-5.838	0.000
INSI	423	1547	1970	584	-5.437	0.000
FINI	448	1522	1970	597	-6.174	0.000
DBI	410	1560	1970	541	-7.478	0.000

The median is used as the test value. The test value is used as a cut point to dichotomize the sample. A run is defined as a sequence of cases on the same side of the cut point. The average number of runs is 431. The number of runs is a measure of randomness. Too many or too few runs, suggest dependence between observations. A lower than expected number of runs indicates the market's over-reaction to information, while higher number of runs reflect a lagged response to information.

The p- value is the probability of obtaining a Z statistic as or more extreme (in absolute value) than the obtained value, if the order of ratings above and below the median is purely random. The negative z-values for all the indices indicate that the actual number

of runs fall short of the expected number of runs. We reject the null hypothesis of random walk when employing the nonparametric assumptions entailed in the runs test. Because the resulting p-values indicate significance of z-values, there is sufficient evidence to conclude that the indices return series are not in random order.

4.3 Unit Root Tests

Table-3 exhibits the results of the three unit root tests, comprising ADF and PP t-statistics and p-values and the KPSS LM-statistic and asymptotic significance. In case of ADF and PP, the null hypothesis of a unit root (non-stationarity) is tested against the alternative of no unit root (stationarity). For the KPSS test, the null hypothesis of no unit root is tested.

The ADF and PP unit root tests rejected the null hypothesis of unit root in all indices, indicating random walk of the return series. The KPSS test results didn't rejected the null hypothesis of no unit root, corroborating the results of the ADF and PP, indicating absence of random walk in Nepalese stock indices.

Table 3: Unit Root Tests Output

Index	ADF		PP		KPSS	
	t-Stat	p-value	t-stat	p-value	KPSS LM-Stat	KPSS Sig.
CBI	-19.6837	0.0000	-35.3623	0.0000	0.1319	--
DBI	-57.1120	0.0000	-55.6947	0.0000	0.1147	--
FINI	-43.3607	0.0000	-68.4390	0.0001	0.0739	--
HOTI	-41.3691	0.0000	-42.2014	0.0000	0.0909	--
INSI	-26.9744	0.0000	-123.7896	0.0001	0.0872	--
MFGI	-26.6270	0.0000	-83.8863	0.0001	0.0236	--
TRDI	-26.6993	0.0000	-159.9217	0.0001	0.0391	--
NEPSEI	-28.8170	0.0000	-33.4324	0.0000	0.1765	--

Notes: For Augmented Dickey-Fuller (ADF) tests hypotheses are H0: unit root, H1: no unit root (stationary). The lag orders in the ADF equations are determined by the significance of the coefficient for the lagged terms. Intercepts and trends in the series. The Phillips-Peron (PP) unit root test hypotheses are H0: unit root, H1: no unit root (stationary). Intercepts and trends in the series. The Kwiatkowski, Phillips, Schmidt and Shin (KPSS) unit root test hypotheses are H0: no unit root (stationary), H1: unit root. The asymptotic critical values for the KPSS LM test statistic at the .10, .05 and .01 levels are 0.3470, 0.4630 and 0.7390 respectively.

We have chosen to estimate an ADF test that includes intercept and trend in the test regression and employs automatic lag length selection using a Schwarz Information Criterion and a maximum lag length of 25. Notice here that the ADF test statistic value is smaller than the critical values so that we reject the null at conventional test sizes. The PP test is similar to ADF tests, but it incorporates an automatic correction to the DF procedure to allow for autocorrelated residuals. In the case of the PP test, the test statistics are more negative than the critical values and hence the null hypotheses of a

unit root in the return series are convincingly rejected. Corroborating with the two tests, the KPSS results also indicate that the Nepalese stock market is not weak form efficient.

4.4 The GARCH (1, 1)

The results of the GARCH (1,1) models estimated for the various indices is given in Table - 4. The models were estimated using the robust method of Bollerslev-Wooldridge's quasi-maximum likelihood estimator assuming Gaussian distribution of the error terms. The Marquardt algorithm was used for the estimation. The coefficients of the AR term of the mean equation are significant

for all models. In the variance equation, the value of the $(\alpha + \beta)$ is higher in all models, which suggests a high persistence of volatility and volatility clustering. A large sum of the coefficients implies that a large positive or a large negative return leads future forecasts of the variance to be high for a protracted period. As is typical of GARCH model estimates for

financial asset returns of developing stock markets, the sum of the coefficients on the lagged squared error and lagged conditional variance is very close to unity in all models implying that shocks to the conditional variance will be highly persistent which is an indication of market inefficiency. These results primarily reject the random walk hypothesis.

Tabel 4: Output of GARCH (1,1) Models

Models	CBI	DBI	FINI	HOTI	INSI	MFGI	TRDI	NEPSEI
Mean Equation								
μ	0.0000 (0.0001)	0.0002 (0.0013)	0.0014 (0.0009)	0.0000 (0.0000)	0.0010 (0.0039)	0.0002 (0.0002)	0.0012 (0.0007)	0.0001 (0.0001)
θ	0.2327* (0.0238)	0.1821** (0.0750)	-0.0731** (0.0606)	-0.4837* (0.0749)	0.1070** (0.2961)	-0.0442* (0.0981)	0.2515* (0.0477)	0.2795* (0.0246)
Variance Equation								
ω	0.0000* (0.0000)	0.0007* (0.0000)	0.0003* (0.0000)	0.0000* (0.0000)	0.0014* (0.0002)	0.0235* (0.0125)	0.0241* (0.0120)	0.0000* (0.0000)
α	0.2181* (0.1753)	0.0862* (0.1160)	0.2573* (0.0653)	0.3113* (0.1019)	0.0382** (0.0000)	0.1455* (0.0396)	0.1254* (0.0854)	0.1845* (0.1614)
β	0.7132* (0.0050)	0.9432* (0.0065)	0.6282** (0.0157)	0.6463* (0.0152)	0.9170** (0.0000)	0.7524* (0.0491)	0.8126* (0.0025)	0.7880* (0.0070)
Adj R-squared	0.0419	-0.1258	0.0479	-0.3086	0.0520	0.0181	0.0254	0.0688
SE of Reg	0.0118	0.0242	0.5530	0.0089	0.0461	0.0153	0.0112	0.0085
SIC	-6.3715	-4.8856	-5.5794	-7.7780	-3.8590	-5.9352	-6.0428	-7.1435
F	22.5376*	27.2640*	25.7434*	25.2350*	27.9795*	10.0590*	8.2540*	37.3734*

* and ** indicate that the coefficient is significant at 1% and 5% level of significance respectively. The numbers in parentheses are standard errors.

The $(\alpha + \beta)$ coefficient of the DBI model is more than one. Such phenomenon is termed as 'non-stationar' in variance. Non-stationarity in variance does not have a strong theoretical motivation for its existence, as would be the case for non-stationarity in the mean

of a price series. The conditional variance forecast will tend to infinity as the forecast horizon increases. The significance of α and β in all models support the hypothesis that conditional volatility changes over times due to volatility clustering. The sum of $(\alpha + \beta)$ a

measure of volatility persistence is very high in most indices which implies that the effects of shocks tend to last for long periods before they die out. The results indicate that returns exhibit persistence and volatility clustering. Weak-form efficiency hypothesis is rejected as it is found that past information helps in predicting future prices.

4.5 The Variance Ratio

The Table-5 reports variance ratios and heteroskedasticity-consistent asymptotically standard normal test statistics in the

parentheses for different aggregation values. For each market index, the variance ratio ratios $VR(q)$ are reported in the main rows, with heteroskedasticity-consistent test statistics $j^*(q)$ given in parentheses immediately below the main row. Under the random walk hypothesis, the value of the variance ratio is one and the test statistics have a standard normal distributions asymptotically. Test statistics marked with asterisks indicate that the corresponding variance ratios are statistically different from one.

Table 5: Variance Ratios for Daily Stock Indices Returns

Indices	Number nq of base observations	Number q of base observations aggregated to form variance ratio			
		2	4	8	16
CBI	1970	1.02 (4.32)*	1.21 (3.91)*	1.48 (3.12)*	2.25 (3.35)*
DBI	1970	1.22 (2.82)*	1.41 (2.94)*	1.75 (3.76)*	1.82 (3.95)*
FINI	1970	1.19 (4.22)*	1.51 (5.14)*	1.67 (4.56)*	1.78 (4.25)*
HOTI	1970	1.21 (3.52)*	1.45 (3.84)*	1.87 (4.26)*	1.83 (4.12)*
INSI	1970	1.52 (5.53)*	1.41 (5.74)*	1.31 (5.60)*	1.22 (5.67)*
MFGI	1970	1.42 (3.22)*	1.51 (3.01)*	1.77 (3.46)*	1.92 (4.05)*
TRDI	1970	1.19 (3.81)*	1.35 (3.87)*	1.48 (3.08)*	1.54 (3.95)*
NEPSEI	1970	1.12 (3.52)*	1.31 (3.94)*	1.47 (4.56)*	1.72 (3.65)*

VR(q) – variance ratio estimate (in the first row) and $j^(q)$ - test statistic (in the second row) for null hypothesis of heteroskedastic increments random walk; the critical value $j^*(q)$ at the 5 percent level of significance is 2.49, asterisk indicates significance at this level; Sampling intervals (q) are in days.*

The variance ratio for the indices is seen normally to increase with q . The estimates of variance ratio are larger than one for all cases and the test statistics indicate that the variance ratios are statistically different from one. The results exhibit that the random walk hypothesis is resoundingly rejected for all market indices.

5. Conclusions

The theoretical and empirical studies of the efficient market hypothesis have made an important contribution to the understanding of the stock market, although the present state of understanding of the issue, especially in the emerging and developing financial markets, is far from being conclusive. The study employed the most comprehensive set of market efficiency tests on stock market indices of different sectors of the Nepal stock exchange and the findings indicate that the Nepalese stock market is weakly inefficient. The absence of random walk in the stock

indices return series infers distortions in asset pricing and risk, a mark of market inefficiency. This provides an opportunity to the traders for predicting the future prices and earning abnormal returns. The implication here is that one should expect a sizeable amount of stock prices on the NEPSE to be either undervalued or overvalued. There is chance of hardworking analysts to consistently outperform the market. The informational efficiency of the market is low. As an institutional policy to improve the capital market, the timely disclosure and dissemination of information to the shareholders and investors on the performance of listed companies should be emphasized. Institutional investment and informed trades should be promoted. It is necessary to gain more insights into the operation and characteristics of the stock market of Nepal in terms of its efficiency to make an informed assessment of the empirical characteristics of the Nepalese stock market.

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